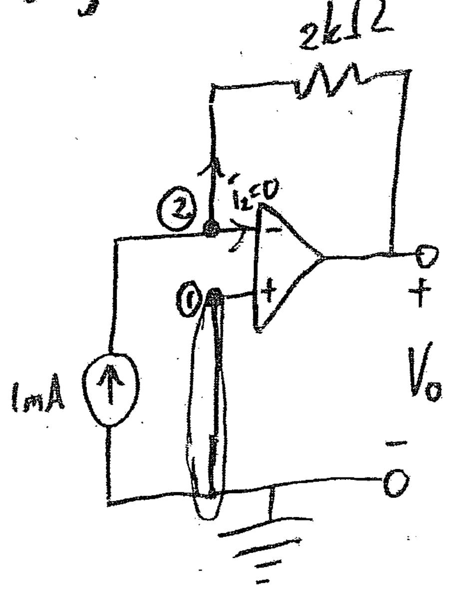
5.8.) a.)



1 $V_{①} = V_{②} = 0V$

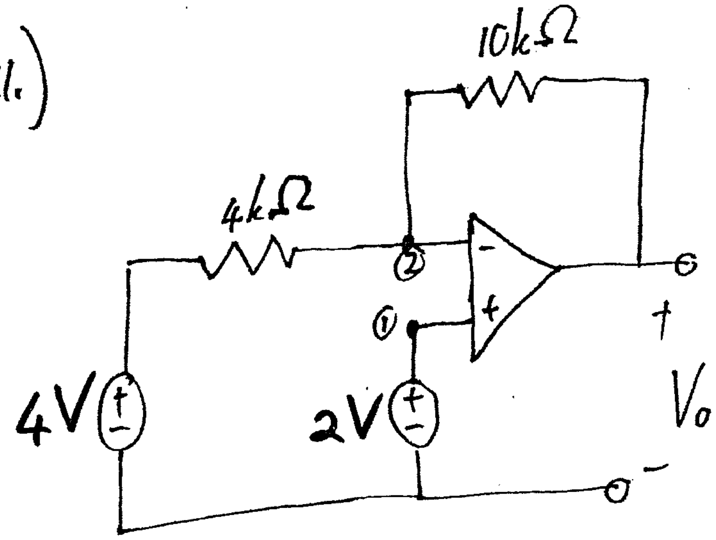
2 KCL at node ②:

$$0{,}001 = \frac{V_2 - V_o}{2000}$$

$$2 = (0) - V_o$$

$$V_o = -2V$$

5.21.)



① $V_1 = V_2 = 2V$

② KCL at node 2:

$$\frac{V_2 - 4}{4000} + \frac{V_2 - V_o}{10\,000} = 0$$

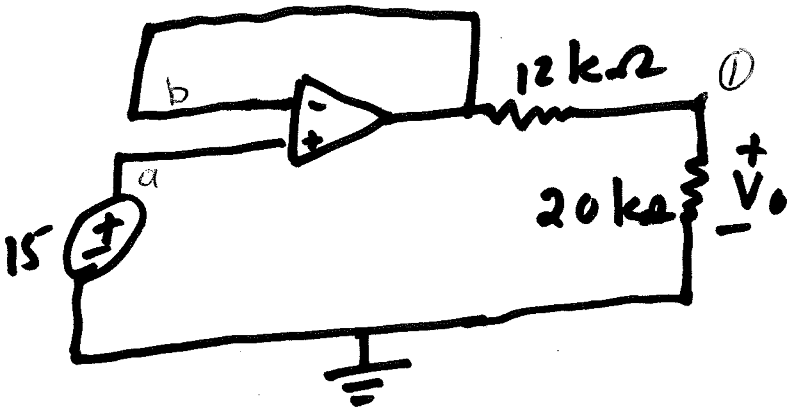
$$\frac{2-4}{4000} + \frac{2 - V_o}{10000} = 0$$

$$-10 + 4 - 2V_o = 0$$

$$-2V_o = 6$$

$$\underline{V_o = -3V}$$

5.25)



Find V_o

$$V_a = V_b = 15\,V$$

KCL and Node 1:

$$\frac{V_b - V_o}{12k} = \frac{V_o}{20k}$$

$$\frac{15 - V_o}{12k} = \frac{V_o}{20k}$$

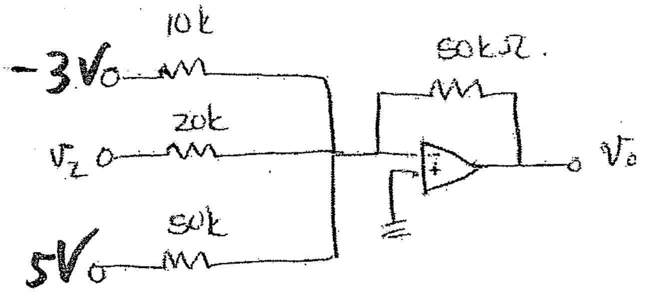
$$(15 - V_o)(5) = 3V_o$$

$$75 - 5V_o = 3V_o$$

$$75 = 8V_o$$

$$\therefore V_o = 9{,}375\,V$$

5.39



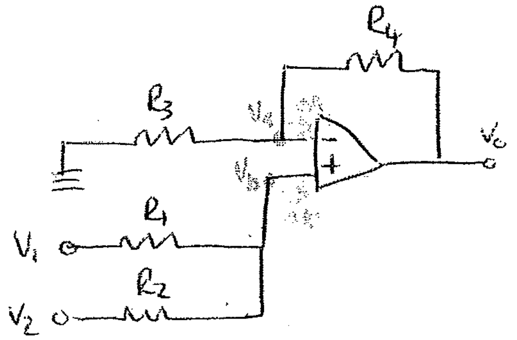
Determine V_2 if $V_o = -16{,}5V$

$$-16{,}5 = -\left[\left(\frac{50}{10}\right)(-3) + \left(\frac{50}{20}\right)V_2 + \left(\frac{50}{50}\right)(5)\right]$$

$$-16{,}5 - 15 + 5 = -2{,}5V_2$$

$$V_2 = 10{,}6V$$

5.44



Show that output voltage is:

$$V_o = \frac{(R_3 + R_4)}{R_3(R_1 + R_2)} (R_2 V_1 + R_1 V_2)$$

* $V_a = V_b$ ——— ①

* KCL @ Va:

$$\frac{0 - V_a}{R_3} = \frac{V_a - V_o}{R_4}$$

$$-V_a = \frac{V_a R_3}{R_4} - \frac{V_o R_3}{R_4}$$

$$-V_a\left(1 + \frac{R_3}{R_4}\right) = -\left(\frac{R_3}{R_4}\right) V_o$$

$$V_a = \left[\frac{\frac{R_3}{R_4}}{1 + (R_3/R_4)}\right] V_o$$

$$= \left[\frac{1}{\left(\frac{R_4}{R_3} + 1\right)}\right] V_o \quad ——— ②$$

* KCL @ Vb

$$\frac{V_1 - V_b}{R_1} + \frac{V_2 - V_b}{R_2} = 0$$

$$R_2 V_1 - R_2 V_b + (R_1 V_2 - R_1 V_b) = 0$$

$$V_b (R_1 + R_2) = (R_2 V_1 + R_1 V_2)$$

$$V_b = \frac{(R_2 V_1 + R_1 V_2)}{R_1 + R_2} \quad ——— ③$$

top.

set $V_a = V_b$.

$$\Longrightarrow \quad \frac{V_o}{\frac{R_4}{R_3} + 1} = \frac{(R_2 V_1 + R_1 V_2)}{R_1 + R_2}.$$

$$V_o = (R_2 V_1 + R_1 V_2)\left[\frac{\left(\frac{R_4}{R_3} + 1\right)^{\times R_3}}{(R_1 + R_2)^{\times R_3}}\right]$$

$$= (R_2 V_1 + R_1 V_2)\left[\frac{R_4 + R_3}{R_3 (R_1 + R_2}\right]$$

Answers

5.8a) $-2V$; b) $V_o = -1V$

5.21) $V_o = -3V$

5.23) $\frac{V_o}{V_s} = -\frac{R_f}{R_1}$

5.33) $i_x = -5mA$; $P_{3k\Omega} = 75mW$

5.37) $V_o = -20,6\,V$

5.39) $V_2 = 10.6\,V$

5.44) Proof

5.47) $V_o = 14,09\,V$